LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Work and Energy

Chapter / Topic: Work and Energy (topic-based)

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SECTION A

Q1. The SI unit of work is the: **[1]**

Answer: (c) joule

Q2. 1 kWh is equal to: **[1]**

Answer: (a) 3.6×10^6 J

Q3. When an object is raised to a height, which form of its energy increases? **[1]**

Answer: (d) Potential energy

Q4. If the velocity of a body is doubled, its kinetic energy becomes: **[1]**

Answer: (b) 4 times ($KE \propto v^2$)

Q5. The work done by a force acting at right angles to the displacement is: **[1]**

Answer: (b) zero

Q6. **Assertion (A):** A man standing still with a heavy suitcase in his hand does no work on it (in the scientific sense). **Reason (R):** Work is done only… **[1]**

Answer: (a)

SECTION B

Q7. Define 1 joule. How much work is done in lifting a 2 kg bag to a height of 1.5 m? **[2]**

Answer: 1 joule is the work done when a force of 1 N moves an object by 1 m in the direction of the force. W = mgh = 2 × 10 × 1.5 = 30 J.

- Definition: 1
- 30 J: 1

OR, answer: Energy is the capacity to do work, measured in joules. Power is the rate of doing work (or using energy), measured in watts (J/s).

- Energy: 1
- Power: 1

Q8. A 60 W bulb is used for 5 hours every day. How much energy does it use in 30 days, in kWh? **[2]**

Answer: E = 60 W × 5 h × 30 = 9000 Wh = 9 kWh (9 units).

- Method: 1
- 9 kWh: 1

SECTION C

Q9. A 2 kg ball is dropped from a height of 5 m. (Ignore air resistance.) **[3]**

(i) PE = mgh = 2 × 10 × 5 = 100 J [1]

(ii) 100 J (all the PE changes into KE). [1]

(iii) ½ × 2 × v^2 = 100, so v^2 = 100 and v = 10 m/s [1]

Q10. Describe the energy changes in (i) a hydroelectric power plant, (ii) a battery-powered torch, (iii) a swinging pendulum. **[3]**

Answer: (i) Potential energy of stored water → kinetic energy of falling water → electrical energy. (ii) Chemical energy in the battery → electrical energy → light (and some heat). (iii) Potential energy at the ends of the swing ↔ kinetic energy at the lowest point, continuously.

- Each: 1

OR, answer: A force F accelerates the object from rest to v over a displacement s. Work done W = F × s = ma × s. From $v^2 - u^2 = 2as$ with u = 0, $s = v^2/2a$. So $W = ma \times v^2/2a = \frac{1}{2}mv^2$. This work is stored as kinetic energy: $KE = \frac{1}{2}mv^2$.

- W = mas: 1
- $s = v^2/2a$: 1
- $KE = \frac{1}{2}mv^2$: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Power is the rate of doing work: P = W/t. SI unit: watt (W). 1 watt is the power of an agent that does 1 joule of work in 1 second. [2]

- Definition + unit: 1
- 1 watt: 1

(ii) Height = 20 × 0.2 = 4 m. W = mgh = 50 × 10 × 4 = 2000 J. P = W/t = 2000/10 = 200 W. [3]

- Height: 1
- Work: 1
- Power: 1

SECTION E

Q12. A pendulum bob of mass 0.2 kg is pulled aside so that it is 0.2 m above its lowest point, and then released. It swings back and forth, but after some… **[4]**

(i) PE = mgh = 0.2 × 10 × 0.2 = 0.4 J [1]

(ii) 0.4 J (all the PE has changed into KE). [1]

(iii) $\frac{1}{2} \times 0.2 \times v^2 = 0.4$, so $v^2 = 4$ and v = 2 m/s. It stops because air resistance and friction at the support slowly change its energy into heat (and sound). Energy is not destroyed, only changed into other forms. [2]

- v = 2 m/s: 1
- Energy converted, not destroyed: 1